

Multi-Modal C-UAS Sensing for Homeland and Mobile Defense

LOE 1 & LOE 2 Proposal



X



DIU Counter UAS Sensing for Homeland and Mobile Defense (PROJ00656)

Interstellar Foundry retains commercial IP for the core sensor-fusion algorithm, while granting the DoD Government Purpose Rights to specific interface APIs and ASCII outputs

Date of Submission: Feb, 26 2026

The Team



- **Company History & Financing:** An early-stage defense technology team founded in January 2026 incubated within the **Georgia Tech Create-X** startup program. Currently operating with pre-seed academic grant funding and institutional prototyping support.
- **Size:** 6 Full-Time Project Members.
- **CAGE Code / UEI:** Pending
- **Key Personnel:**
 - Parth Patel (Mechanical Engineer)
 - Calkin Garg (Computer Science)
 - Zheng-Yin Lee (Electrical/Computer Engineer)
 - Ethan Yee (Mechanical Engineer)
 - Carlos Cordova (Electrical/Computer Engineer)
 - Rashod Abdurasulov (Computer Science)

The Problem & The DIU Mandate

"Today, autonomy thrives; tomorrow, intelligence prevails."

The Capability Gap

- > Legacy sensor suites lack the sensitivity to detect low **Radar Cross Section (RCS)** profiles of modern sUAS.

The Cost Barrier

- > Existing systems possess a sustainment burden preventing the scale necessary to protect dispersed infrastructure.

Our Direct Solution

- > A sub-\$1000 modular ground station utilizing Commercial Off-The-Shelf (COTS) edge computing to deliver high-fidelity target tracking.



System Architecture



1

**Primary
(Radar) :**

Microwave Doppler/FMCW
radar establishes initial range,
velocity, and detection triggers.

2

**Secondary
(Optical)**

Cues a high-fidelity camera to
achieve precise spatial tracking
and object classification.

3

**Edge Compute
Engine**

Powered by an **NVIDIA Jetson**
to handle localized, low-latency
AI processing and sensor data
ingestion without any cloud
compute dependencies.

4

The Handshake

A **Kalman filter** processed on
the Jetson meshes the radar and
optical feeds to eliminate false
positives (birds, weather, ground
clutter) and lock onto legitimate
Group 1 and 2 targets

LOE Compliance

LOE 1 Compliance: Homeland Defense

Data Output:

Delivers target data and required **Radar Cross Section (RCS)** metrics in the mandated **ASCII format**.

Fire Control Integration

Engineered for rapid, plug-and-play integration with designated C-UAS command and control systems.

Power

Fully compatible with standard U.S. military generators (5 kW-30 kW) and shore power.

LOE 2 Compliance: Tactical & Mobile

GNSS-Denied Operations

Software utilizes OpenCV, Visual Odometry (ORB), and SLAM mapping to localize threats relative to a local origin, ensuring resilience when GPS is jammed.

Platform Integration

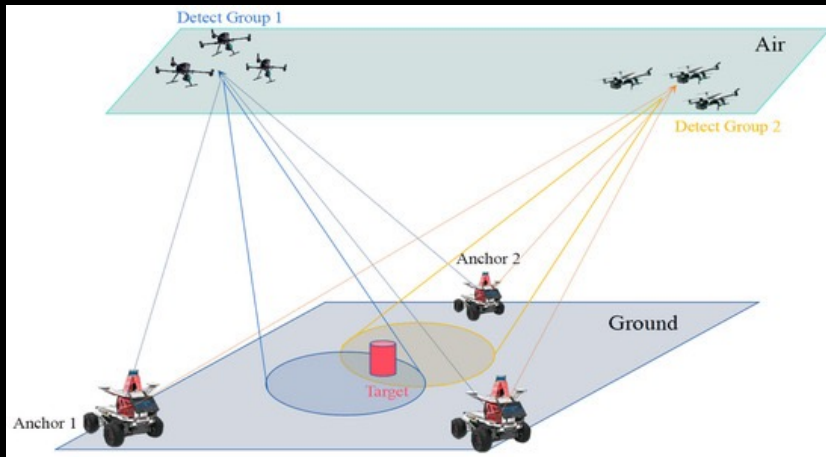
Compact form factor natively supports mounting on standard tactical vehicles (ISV, JLTV, FMTV, HEMTT).

Passive Posture

Engineered with low physical and spectral characteristics for maximum survivability.

Meeting the DIU Technical Thresholds

Detection Range Evolution	—>	Detection Range Evolution: MVP demonstrates sub-100m detection to validate the Kalman filter; scaled production pathways utilize upgraded COTS radars to meet the DIU >2 km target for Group 1 UAS.
Inner Perimeter Defense	—>	Inner Perimeter Defense: System is optimized for the 50m to 500m range, capturing threats that slip past macro-radar installations.
Low-Altitude & Hover Capabilities	—>	Low-Altitude & Hover Capabilities: Designed for high fidelity at <50m AGL (Above Ground Level) and capable of tracking hovering UAS exhibiting 0 Hz Doppler signatures.
Track Limits	—>	Track Limits: Capable of simultaneous multi-target tracking with high refresh rates and automatic track separation.



Hardware Subsystems

Radar (Tiered Scaling) —>

Optical Suite —>

Edge Processing —>

Tactical Footprint



From **mmWave 24GHz** (~\$75)
for MVP algorithm
validation, scaling to
TI IWR6843AOPEVM (~\$220)
for high-speed DSP and
>50m tracking.

Highly compatible with
OAK-D Pro (RGB/stereo
depth) or **Hikvision**
Ultra (8MP motorized
target locking).

NVIDIA Jetson Nano delivers
localized, cloud-independent AI
(60+ TOPS) on a highly efficient
7-25W power draw.

Sub-10 lb modular design
supports under 15-minute
deployment on ISV, JLTV,
FMTV, and HEMTT platforms.

Software Subsystems

GNSS-Denied Navigation —>

Spectral
Stealth —>

Cybersecurity



Converts 2D pixels to 3D map
coordinates using OpenCV and
SLAM to localize threats
without GPS.

Air-gapped serial communication
and passive optics ensure maximum
EMI robustness and low
physical/spectral signatures.

Architecture strictly
complies with **DOD**
Instruction 8510.01 (RMF) to
streamline the Authorization
to Operate (ATO).

Phased Development and ROM Pricing

Development and Manufacturing

COTS Advantage	—>	Bypasses expensive proprietary defense foundries by leveraging highly available commercial supply chains used in industrial robotics.
Subsidized R&D:	—>	Leverages Georgia Tech Create-X facilities to drastically reduce prototyping, fabrication, and initial tooling costs.
Evolution	—>	MVP phase validates the Kalman filter logic; Phase 2 transitions hardware into ruggedized, MIL-SPEC for extreme weather operations.

Tiered ROM Pricing Strategy

Tier	Tier 1	Tier 2	Tier 3
Units	1-5	6-25	26-100+
Key Deliverables	MVP hardware for rapid field-envelope testing at approximately \$500-\$1,000 per node.	Low-Rate Initial Production (LRIP) scaling with streamlined assembly to further reduce per-unit costs.	Mass manufacturing achieves economies of scale, allowing for a dispersed sensing layer at a fraction of legacy system costs.

Tactical Performance Metrics & Business Viability

Sensing Envelope & Accuracy:

- **Search & Track Coverage:** 360° azimuth and 30° elevation search coverage; up to 90° elevation in track coverage. Capable of steering search/track below 0° for elevated vehicle emplacements.
- **Update Rates & Error Metrics:** System operates at a **10Hz** track update rate with an estimated azimuth/elevation error of **< 5 mrad**, easily supporting downstream effector cueing.
- **Tactical Mobility:** Maintains effective sensor fusion and track continuity at vehicle speeds up to 50mph

Emissions & C2 Integration:

- **Active Sensor Profile:** The primary radar utilizes the **24 GHz FMCW** (K-band) spectrum. Emissions are strictly limited 20 dBm to ensure a low spectral signature and prevent enemy targeting.
- **C2 Integration List:** Architecture natively outputs target data via ASCII and standard APIs designed for immediate integration with **FAAD C2** (Forward Area Air Defense Command and Control), pending Phase 2 DoD system access.

Company Viability & Funding:

- **Financial Runway:** Currently operating on **\$1000-\$5000** in pre-seed academic grant funding backed by the Georgia Tech Create-X program.
- **Corporate Structure:** Operating with zero external debt and fully retained equity, prioritizing rapid, lean prototyping over bloated R&D cycles.

Strategic Value Proposition

"Peace cannot be kept by force; it can only be achieved by understanding." — Einstein

Current State & Maturation: Currently at **TRL 4** (Bench/Lab validated sensor logic). Aggressively moving to **TRL 6-7** by integrating the full stack for active field testing.

Phase 2 Execution: Fully prepared to execute live demonstrations and supply the mandated **DD Form 1494** within 30 days of notification.

Call to Action: Ready to demonstrate how resilient, scalable, and autonomous COTS hardware can provide high-fidelity C-UAS situational awareness.

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